CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 2: **Acids, Bases and Salts** | Quick revision, one page

Indicators and pH

- Acid: turns blue litmus red; base: turns red litmus blue; phenolphthalein is pink in base and colourless in acid; methyl orange is red in acid, yellow in base.
- pH scale 0-14: pH < 7 acidic, pH = 7 neutral, pH > 7 basic. Lower pH = stronger acid; higher pH = stronger base.
- Acid + water: $\mathrm{HCl} \longrightarrow \mathrm{H^+ + Cl^-}$; $\mathrm{H^+}$ exists as $\mathrm{H_3O^+}$. Always add acid to water, never water to acid.

Reactions of acids and bases

- Acid + metal $\rightarrow$ salt + $\mathrm{H_2}$: $\mathrm{Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2}$ (gas burns with a pop sound).
- Acid + metal carbonate / hydrogencarbonate $\rightarrow$ salt + $\mathrm{CO_2}$ + $\mathrm{H_2O}$:
  $\mathrm{Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2}$.
- Base + metal (e.g. Zn): $\mathrm{Zn + 2\,NaOH \longrightarrow Na_2ZnO_2 + H_2}$. Base + non-metal oxide $\rightarrow$ salt + water.
- Neutralisation: acid + base $\rightarrow$ salt + water, e.g. $\mathrm{NaOH + HCl \longrightarrow NaCl + H_2O}$.
- Lime water test: $\mathrm{Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O}$ (milky); excess $\mathrm{CO_2}$ gives soluble $\mathrm{Ca(HCO_3)_2}$.

Common salts

- Common salt: NaCl. Chlor-alkali: $\mathrm{2\,NaCl + 2\,H_2O \longrightarrow 2\,NaOH + Cl_2 + H_2}$ (anode $\mathrm{Cl_2}$, cathode $\mathrm{H_2}$).
- Bleaching powder: $\mathrm{Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O}$; uses: bleaching, disinfecting water.
- Baking soda: $\mathrm{NaHCO_3}$, from $\mathrm{NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3}$; heating:
  $\mathrm{2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2}$.
- Washing soda: $\mathrm{Na_2CO_3 \cdot 10\,H_2O}$ (from $\mathrm{Na_2CO_3}$ by recrystallisation).
- Plaster of Paris: $\mathrm{CaSO_4 \cdot \tfrac{1}{2}\,H_2O}$ from $\mathrm{CaSO_4 \cdot 2\,H_2O}$ (gypsum) heated at 373 K;
  $\mathrm{CaSO_4 \cdot \tfrac{1}{2}\,H_2O + \tfrac{3}{2}\,H_2O \longrightarrow CaSO_4 \cdot 2\,H_2O}$.

Everyday applications

- Antacids (mild bases, e.g. $\mathrm{Mg(OH)_2}$) relieve indigestion; tooth decay begins at pH < 5.5; acid rain pH < 5.6.
- Water of crystallisation: fixed number of water molecules in a salt crystal (blue $\mathrm{CuSO_4 \cdot 5\,H_2O}$ becomes white on heating).

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